

Statistical Hypothesis Testing



Dr. Mohammad Rafiqul Islam
Associate Professor of Statistics
Dept. of Mathematics and Natural Sciences (MNS)
BRAC University, Dhaka, Bangladesh
mrafiq@bracu.ac.bd

Statistical Hypothesis Testing

- Statistical Hypothesis testing is a method of making statistical inference from the sample (or experimental) data
- That is hypothesis testing is concerned with making decisions using data
- Hypothesis tests are used to test the validity of a claim that is made about a population

Statistical Hypothesis Testing

The formulation and philosophy of hypothesis testing is created in 1915–1933 by 3 men:

- Ronald A. Fisher (1890-1962), British statistician
- Jerzy Neyman (1894-1981), Polish-American mathematician and statistician
- Egon S. Pearson (1895-1980), British statistician

Hypothesis Testing

Hypothesis testing is statistical inference concerned with

- The question of interest
- Collecting data (called sample data)
- Then examining data
- Making decisions using data

In a formal hypothesis test, hypotheses are always statements about the population.

Introduction

- Statistical hypothesis testing is used to determine whether the sample (or an experiment) provides enough evidence to reject or accept a proposition (proposal).
- For example, suppose you want to verify the statement that smoking causes lung cancer. We use statistical hypothesis testing.
- In this case, we will first assume a hypothesis that smoking causes lung cancer. This is called the 'null hypothesis', which is central to any statistical hypothesis testing.
- Then we will take a sample and
- based on the sample data we will come to a conclusion where smoking causes lung cancer through some statistical procedures.

Let us formulize the steps of Hypothesis Testing

Steps for Hypothesis Testing

1. Choose the null and alternative hypotheses
2. Choose a significance level
3. Calculate the test statistic
4. Calculate critical region (or the p-value)
5. State the conclusion

1. Formulation of the null and alternative hypotheses

- ***Null hypothesis:*** The null hypothesis is a statement that you want to test.
- A null hypothesis is usually labeled by H_0 .
- The null hypothesis is assumed that the statement is true and statistical evidence is required to reject it in favor of a research
- ***Alternative hypothesis:*** The alternative hypothesis is different from the null hypothesis that is the null hypothesis is not true.
- It is denoted by H_A or H_1

For example, if we want to check that the average income of inhabitants living in a city is 800\$ that is the null hypothesis is

H_0 : the average income in a city is 800 \$

(**H_0** : $\mu=800\$$)

And the alternative hypothesis H_A is

H_A : the average income in a city is not 800 \$

(**H_A** : $\mu \neq 800 \$$)

Give another example of null and alternative hypotheses

(Say on an average sleeping hours of students is 5 hours)

8 ## Think about 5 examples of null and alternative hypotheses

2. Specification of the level of significance:

- The Level of Significance denoted by α is pre-determined before taking the sample.
- It does not depend on the sample at all. It is the probability at which we consider something unusual (error).
- The most commonly used significance levels are 0.05 (5%) and 0.01 (1%).
- With a significance level of 0.05, you have a 5 percent chance of rejecting the null hypothesis, even if it is true.
- In other words, at the 0.05 level of significance, based on the sample result, the decision to reject the hypothesis may be incorrect 5% of the times.
- Similarly, at (1%) 0.01 level of significance, based on the sample result, the decision to reject the hypothesis may be incorrect 1% of the times
- You must choose your significance level before conducting the hypothesis tests.

3. Decision for a suitable statistical test(also called test statistic):

A test statistic)is calculated from the observed data and this forms the basis of the statistical test. The most frequently test statistics are normal test(Z test). t test, F test and Chi square test.

4. Calculate critical region (or the p-value) :

(i) calculation of the critical region (tabulated value):

The critical region is calculated from the tabulated value of the test statistics.

Critical region depending on alternative hypothesis

(ii) **p-value** : As most of the statistical software (R, SPSS, Python) provides the P-value

We can use any either critical region or p-value; or we can use both of them..

5. Statistical decision and Interpretation of the test result:

1. "Critical value" (or "critical region" or "rejection region") approach: In this approach calculated value of test statistics is compared with the tabulated value of the test statistics (called critical region or critical value).

- ❖ If calculated value of test statistic greater than tabulated value (fall in the critical region) we reject the null hypothesis.

(2) "P-value" approach: In the p value approach:

- ❖ If p-value < 0.05 leads to rejection of the null hypothesis and acceptance of the alternative hypothesis at 5% level of significance. (If $p \geq 0.05$, the decision is "fail to reject the null hypothesis")
- ❖ If p-value < 0.01 leads to rejection of the null hypothesis and acceptance of the alternative hypothesis at 1% level of significance

Two criteria for a good test

Significance level: We know that the level of significance is the probability of rejecting the null hypothesis H_0 when it is in fact true. Most commonly used level of significance are **5% and 1%**.

Type I error : The error that we are doing by rejecting the true null hypothesis is called type I error and its probability is denoted by α (significance level) .

Type II Error – The error that we are doing by failing to reject (that is accept) null hypothesis, H_0 when it is actually false is known as type II error and its probability is denoted by β

- **Power:** Power is $1 - \beta$. So power is the Probability of reject a H_0 when it is truly

Criteria for a test

	Truth about the population	
	H_0 is true	H_0 is false
Decision based on sample(hypothesis test)		
Accept H_0	Correct Decision (probability = $1 - \alpha$)	Type II Error - fail to reject H_0 when it is false (probability = β)
Reject H_0	Type I Error - rejecting H_0 when it is true (probability = α)	Correct Decision (probability = $1 - \beta$)

Example of type I and type II error

To understand the interrelationship between type I and type II error, and to determine which error has more severe consequences, consider the following example.

A medical researcher wants to compare the effectiveness of two medications.

The null and alternative hypotheses are:.

Null hypothesis : The two medications are equally effective

$$H_0: \mu_1 = \mu_2$$

Alternative hypothesis: The two medications are not equally effective.

$$H_A: \mu_1 \neq \mu_2$$

H_0 is true: A type I error occurs if the researcher rejects the null hypothesis and concludes that the two medications are different when, in fact, they are not. If the medications have the same effectiveness, the researcher may not consider this error too severe because the patients still benefit from the same level of effectiveness regardless of which medicine they take.

H_0 is false: However, if a type II error occurs, the researcher fails to reject the null hypothesis when it is false. That is, the researcher concludes that the medications are the same when, in fact, they are different. This error is potentially life-threatening if the less-effective medication is sold to the public instead of the more effective one.

As you conduct your hypothesis tests, consider the risks of making type I and type II errors.

If the consequences of making one type of error are more severe or costly than making the other type of error, then choose a level of significance and a power for the test that will reflect the relative severity of those consequences.

Statistical significant

*When we reject the null hypothesis and accept the alternative, the test is known as **statistically significant***

- **if** we reject the null hypothesis at 5% level of significance result is called **statistically significant**

- **Conventional definition of “very small”:**

If we reject the null hypothesis at 1% level of significance result is called **highly statistically significant**

P-value (Probability Value)

- **P –value** stands for probability value(numbers between 0 and 1)
- **P –value is the** of probability of getting the value that is either the same or more extreme than the actual observed value from the sample by chance (or randomly) if the null hypothesis is true.
- Thus, the smaller p-value means there is the less evidence of getting the sample result by chance (or randomly) and thus less chance of rejecting the alternative hypothesis H_A , or
- In other words, the smaller the p-value, the stronger the sample evidence is for rejecting H_0 .

P-value (Probability Value)-example

- If the significance level is fixed at, $\alpha = 5\% = 0.05$.
- Based on test statistics p-value is obtained say, $p = 0.03$
- Our p-value which is less than 0.05. ($p < 0.05$)
- We will reject the null hypothesis H_0 as p-value is less than 0.05
- The p-value 0.03 means there is 3% probability of getting the sample result by chance
- that is 3% chance (< 0.05) of obtaining sample result mistakenly.
- So we reject the null hypothesis H_0 and accept the alternative
- In other words, the p-value is a measure of the strength of the evidence in your data against H_0 . Usually, the smaller the p-value, the stronger the sample evidence is for rejecting H_0 .

summarize- of P-value (Probability Value)

- p-values are probabilities (numbers between 0 and 1)
- A small p-value (typically ≤ 0.05) indicates strong evidence against the null hypothesis, so you reject the null hypothesis.
- A large p-value (> 0.05) indicates weak evidence against the null hypothesis, so you fail to reject the null hypothesis.
- p-values very close to the cutoff (0.05) are considered to be marginal (could go either way). Always report the p-value so your readers can draw their own conclusions.
- Note that is, a p-value of 0.06 , we fail to rejected hull hypothesis at an 0.05 level of significance, as $p\text{-value} > 0.05$. test is not significant
- but it would be rejected at an 0.10 level of significance. As $p\text{-value}(0.06) < 0.10$. test is significant

For example, suppose a grocery store claims that the delivery time of online order on an average is 40 minutes.

To verify the claim, you conduct a hypothesis test because you believe the null hypothesis, H_0 , that the mean delivery time is 40 minutes is correct. Your alternative hypothesis (H_A) is that the mean time is greater than 40 minutes. You randomly took a sample of some delivery times and run the data through the hypothesis test, and you got the p-value based on the test 0.002, which is much less than 0.05.

You will reject the null hypothesis as this p-value 0.002 is less than 0.05, you conclude that the claim of the grocery store is wrong.

P-value (Probability Value)

We can get the most common p-values using the distribution calculator as following link:

https://bitly.com/dist_calc

One-tailed test / two-tailed test

The null hypothesis is $H_0: \mu = \mu_0$ and the alternative hypothesis is one of the following:

$$H_A: \mu \neq \mu_0$$

or

$$H_A: \mu < \mu_0$$

or

$$H_A: \mu > \mu_0$$

(Two-tailed)

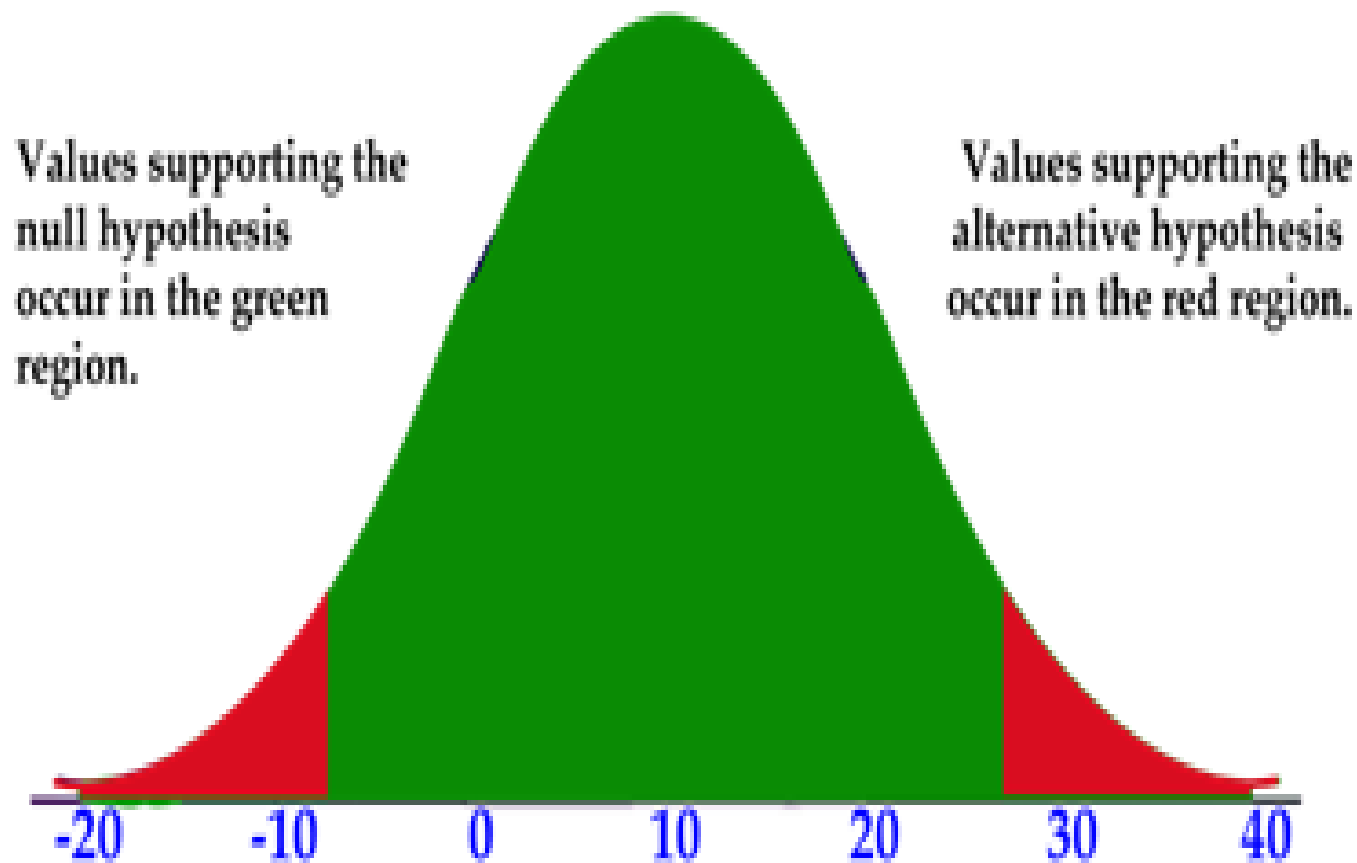
(Left-tailed)

(Right-tailed)

(one-tailed)

(one-tailed)

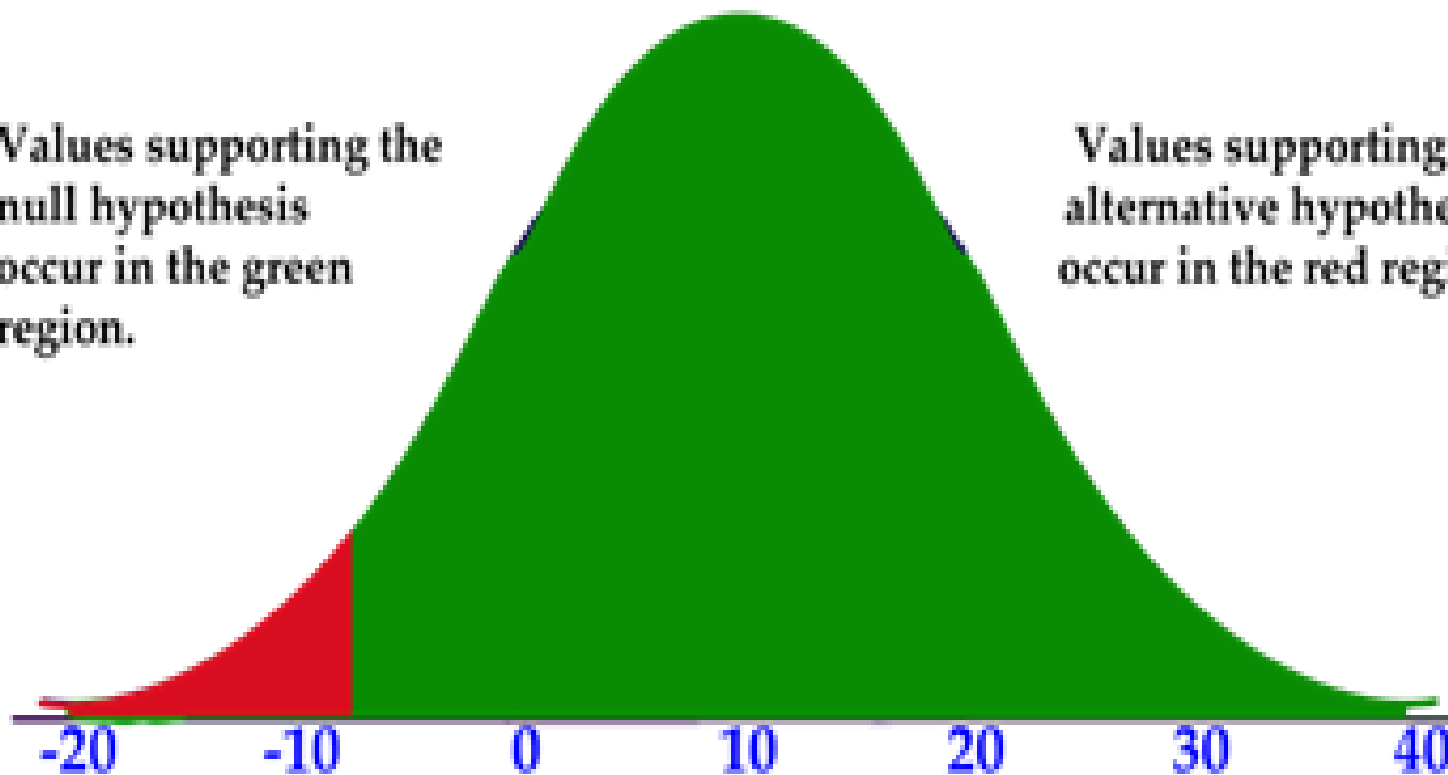
Two-tailed test ($H_A : \mu \neq \mu_0$)



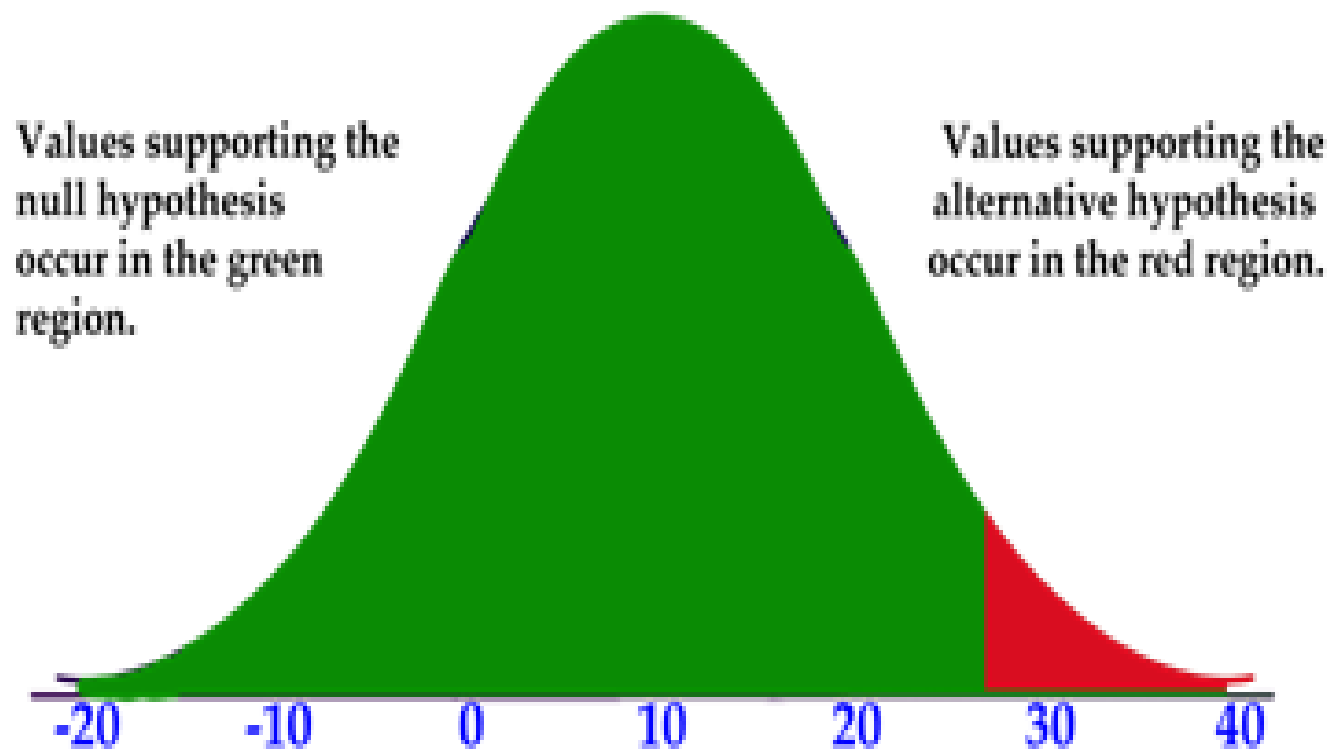
Left-tailed test($H_A : \mu < \mu_0$)

Values supporting the null hypothesis occur in the green region.

Values supporting the alternative hypothesis occur in the red region.



Right-tailed test($H_A : \mu > \mu_0$)



■ Test Statistics:

The most commonly used statistical tests are:

- Normal or Z test
- t test
- F test
- Chi square test (X^2)

Conditions for Hypothesis tests

Independence: Sampled observations must be independent (random sample/assignment & if sampling without replacement, $n < 10\%$ of population)

Sample size/skew: sample size $n \geq 30$, larger if the population distribution is very skewed.